The Rise of Agentic AI: From Generative Models to Autonomus Agents

Dr. Sunil Saumya, Assistant Professor

Dept of Data Science & Artificial Intelligence
Indian Institute of Information Technology Dharwad

sunilsaumya @iiitdwd.ac.in



August 10, 2026

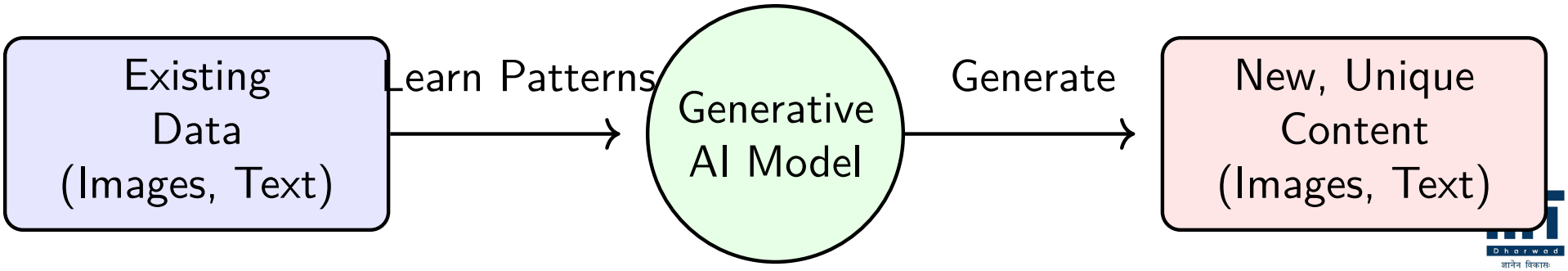
Outline

1. Generative AI Introduction
   - Tradition AI Vs Generative AI
2. Use-case: Hiring a Bakend Engineering
   - Chatbot V1: Gen AI Chatbot
   - Chatbot V2: RAG based Chatbot
   - Chatbot V3: Tool-Augmented Chatbot
   - Chatbot V4: Agentic Chatbot
3. Defining Agentic AI
   - Agentic AI Characteristics
   - Agentic AI Components

Generative AI Introduction

What is Generative AI?

- **Generative AI** is a category of artificial intelligence that can **generate new and unique content**.
- Unlike traditional AI models that are trained to classify or predict based on existing data, generative models are trained to **learn the underlying patterns and structures** of their training data.
- This allows them to create new outputs that are similar to, but not a direct copy of, what they were trained on.
- Common forms of content generated by these models include:
  - **Text**: Creating articles, stories, and summaries, Example: *ChatGPT*.
  - **Images**: Generating art, photorealistic pictures, Example: *Diffusion Models*.
  - **Audio**: Synthesizing music, sound effects, Example: *ElevenLabs*.

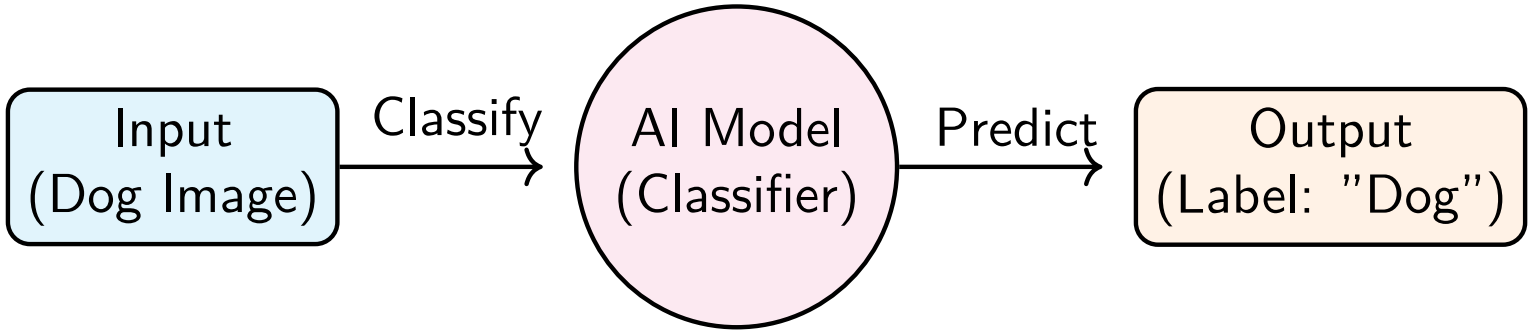


Tradition AI Vs Generative AI

Traditional AI vs Generative AI

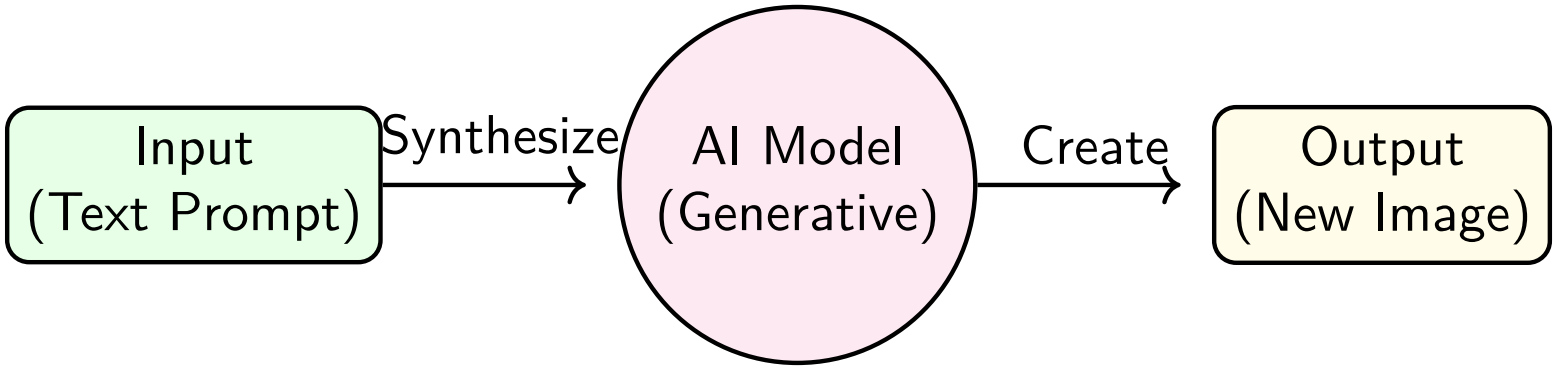
Traditional AI (Discriminative)

- Focuses on classification and prediction.
- Distinguishes between existing categories.



Generative AI

- Focuses on creating new content.
- Learns patterns to create from scratch.



Application Areas of Generative AI



Content Creation
Generating articles, summaries, code, and marketing materials.



Creative Arts
Producing new images, music, and videos from text prompts (e.g., Midjourney, Stable Diffusion).



Customer Service
Powering intelligent chatbots and virtual assistants for personalized support.



Scientific Research
Accelerating drug discovery by generating new molecular structures.



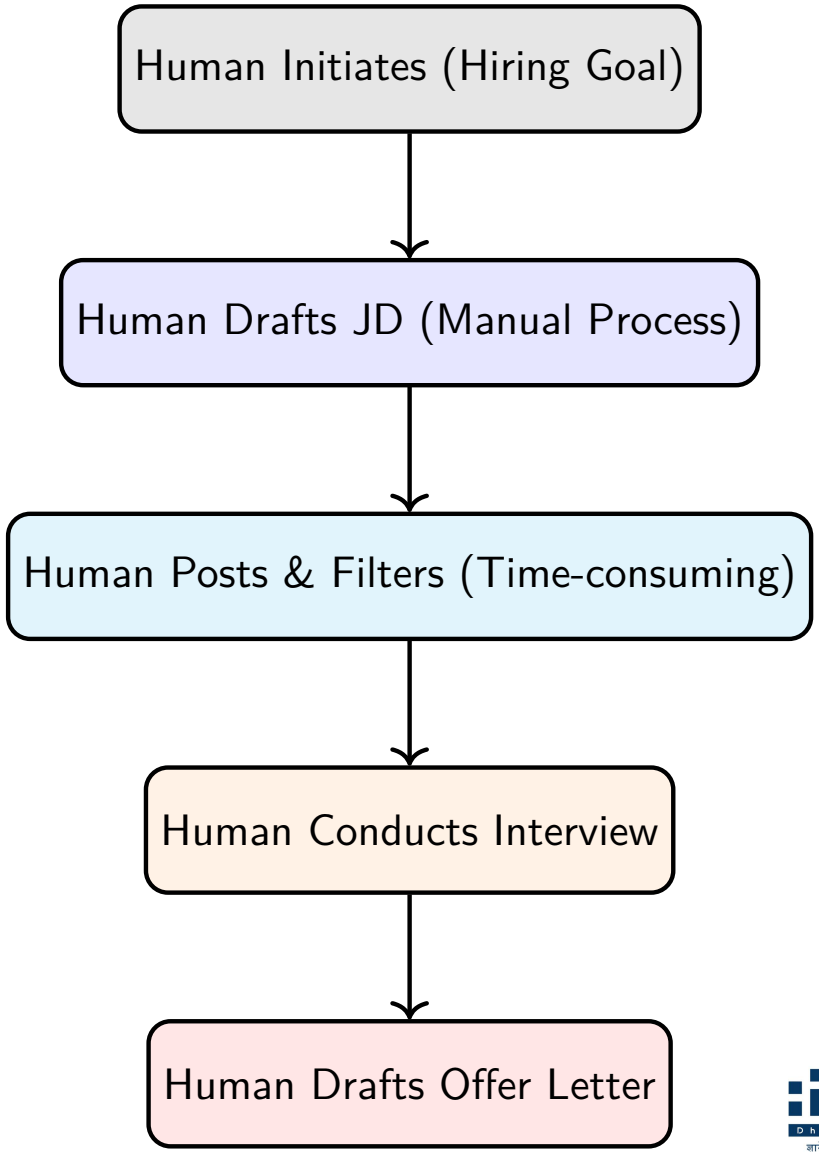
Software Development
Assisting with code generation, bug fixing, and automated testing.

Use-case: Hiring a Bakend Engineering

A Traditional Scenario: Hiring a Backend Engineer

Goal: To hire a backend engineer.

- **Job Description:** A hiring manager or recruiter manually drafts the JD from scratch.
- **Posting & Shortlisting:** The recruiter manually posts to multiple sites and then spends hours reviewing hundreds of resumes.
- **Interview:** The team creates a new set of interview questions, often repeating similar ones.
- **Offer & Onboarding:** The HR department manually creates and sends offer letters and onboarding documents.



Chatbot V1: Gen AI Chatbot

Hiring with a Generative AI Chatbot

User: I want to hire backent engineering with 2-4 years of expeience.

Gemini: Ok, how can I help you?

User: Can you please hepl me draft a Job Description (JD)?

Drafting a Job Description with Generative AI

User: I want to hire backent engineering with 2-4 years of ex-peience.

Gemini: Ok, how can I help you?

User: Can you please hepl me draft a Job Description (JD)?

Generated Job Description

Backend Engineer (2-4 years experience) Responsibilities:

- Design, develop, and maintain backend services and APIs.
- Write clean, maintainable, and efficient code in Python.

Requirements:

- Bachelor's degree in Computer Science or a related field.
- 2-4 years of professional experience in backend development.
- Strong proficiency in Python, Django/Flask, and SQL/NoSQL databases.

Hiring with a Generative AI Chatbot

User: Where should I post this?

Gemini: You can post it on LinkedIn and YouTube.

User: Thank you, I will do that.

Now, the user will manually post the JD to these platforms.





Shortlisting Candidates with Generative AI Chatbot

User: I have received 10 applications. Can you please guide me in shortlisting?

Gemini: Look for relevant skill, minimum experience, team work ability.

User: Thank for guiding, I will do the rest.

Shortlisted Candidates

User: I have received 10 applications. Can you please guide me in shortlisting?

Gemini: Look for relevant skill, minimum experience, team work ability.

User: Thank for guiding, I will do the rest.

Applications Review

Candidate	Status
Application 1	Shortlisted
Application 2	Rejected
Application 3	Shortlisted
Application 4	Rejected
Application 5	Shortlisted

Scheduling Interviews with a Generative AI Chatbot

User: I want to schedule interview with 5 candidates. Can you please draft me an email.

Gemini: Sure, Drafting an email.

User: Thanks, I will mail them.

Drafted email by Generative AI Chatbot

User: I want to schedule interview with 5 candidates. Can you please draft me an email.

Gemini: Sure, Drafting an email.

User: Thanks, I will mail them.

Generated Email Draft

Subject: Interview Invitation for Backend Engineer Position

Dear [Candidate Name], Thank you for your interest in the Backend Engineer position at [Company Name]. We were impressed with your application and would like to invite you to a virtual interview to discuss your experience. Please reply to this email to confirm your availability. We look forward to speaking with you soon.

Best regards, [Your Name]

Interview Questions Generation using Chatbot

User: Can you generate some interview questions for a backend engineer?

Gemini: Sure. Please provide some key skills you'd like to focus on to make the questions more relevant.

User: Let's focus on Python, Django, and databases based on our earlier JD.

Generated Interview Questions

Gemini: Understood. Generating a set of questions for you now.

User: Thanks, these look great. I'll use them.

Generated Interview Questions

Technical Skills:

- Explain the difference between Django and Flask.
- How would you optimize a slow SQL query?
- What is an ORM, and what are its pros and cons?

Problem-Solving:

- Describe a complex problem you solved in a previous project.
- How would you design a simple REST API for a blog application?

Behavioral:

- Tell me about a time you had a conflict with a teammate and how you resolved it.

Drafting an Offer Letter with Chatbot

User: We've selected a candidate. Can you help me draft an offer letter?

Gemini: Absolutely. Please provide the candidate's name, start date, and salary.

User: Candidate is Mr. Rahul, with a start date of 1st Oct and salary of 95,000 Rupees per Month.

Gemini: Understood. Drafting the offer letter for you now.

Generated Offer Letter

Gemini: Your offer letter for Mr. Rahul has been drafted.

User: Thank you! This is very helpful. I'll send it now.

Generated Documents

Offer Letter: Mr. Rahul

Dear Mr. Rahul,
We are delighted to offer you the position of Backend Engineer at [Company Name]. Your employment is scheduled to begin on **October 1st**, with a salary of **Rs. 95,000 per month**.
Please review the full offer details in the attached document. We look forward to having you join our team. Sincerely,

[Your Name]

Automating Onboarding with Chatbot

User: Rahul has accepted the offer. Can you help me with the onboarding process?

Gemini: Congratulations! I can prepare a personalized onboarding checklist and welcome message for him.

User: Perfect. Please go ahead and prepare the documents.

Automated Onboarding Materials

Gemini: The personalized welcome message and onboarding checklist for Rahul are ready.

User: Perfect. This saves a lot of time. Thank you!

Generated Onboarding Documents

Dear Rahul,
Welcome to [Company Name]! We are thrilled to have you join us as a Back-end Engineer. We've prepared a checklist to help you get started on your first day, October 1st.

Onboarding Checklist

- Complete HR paperwork and submit ID
- Review employee handbook
- Set up workspace and access credentials
- Meet with your team lead, [Manager's Name]

Advantages of Using a Chatbot in the Hiring Process

- **Significant Time Savings**: The chatbot drastically reduces the time spent on every stage, from drafting job descriptions and interview questions to generating offer letters and onboarding documents.
- **Process Consistency**: Ensures a uniform and professional standard for all communications and documents throughout the hiring pipeline.
- **Enhanced Scalability**: Allows a small HR team to manage a high volume of candidates efficiently, making it ideal for rapid company growth.
- **Improved Focus**: Automates tedious, repetitive tasks, freeing up recruiters and hiring managers to concentrate on high-value activities like personal outreach, candidate engagement, and final decision-making.

Limitations of Chatbot

- **Reactive, Not Proactive:** The chatbot is a tool that requires continuous human prompts to operate. It cannot act autonomously or take initiative.
- **Limited Context and Memory:** It lacks a persistent memory of the entire hiring project. Each interaction is a new query, requiring the user to manually provide context for every task.
- **Generic Output:** Without integration with your company's specific data, the chatbot generates generic content (e.g., job descriptions, interview questions) that a human must still refine.
- **Can't Take Action:** The chatbot can only provide text-based responses. It cannot perform real-world actions like sending emails, scheduling meetings, or posting on social media platforms.

Solved vs. Unsolved Limitations: Generic Chatbot V1

Problem	Status
Reactive, Not Proactive	×
Lacks Memory	×
Specific Advice	×
Can't Take Action	×

Chatbot V2: RAG based Chatbot

Introducing a Grounded Chatbot: Version 2

Problem: The previous chatbot was reactive and provided generic advice because it lacked access to our company's specific data.

Solution: By connecting the chatbot to our internal knowledge base, we can provide it with the context needed to give personalized and accurate advice.

This new version addresses two key limitations:

- **No Memory:** It is now "grounded" in our company's data.
- **Generic Advice:** Its responses are now tailored to our specific needs.

The Grounding Data for our Chatbot

JD Templates

- Past JDs used by the company
- Example of high-performing JDs
- Variations for remote vs in-office, junior vs senior

Hiring Strategies

- Best platform for hiring
- Best practices for hiring
- Internal salary band
- Internal checklist to hire for various profiles
- Interview question banks for various profiles

Onboarding Checklist

- Offer letter template
- Welcome email template
- Employee policies

RAG Chatbot V2: Drafting a JD with a RAG

User: Can you please help me draft a Job Description to hire a Backend Engineer?

Gemini: Yes. I will draft a JD using the company's past templates and high-performing examples to ensure it aligns with our standards.

Backend Engineer (2-4 years experience)

JD based on high-performing template #3

Responsibilities:

- Design, develop, and maintain efficient backend services and APIs.
- Collaborate with cross-functional teams.

Requirements:

- 2-4 years of experience.
- Strong proficiency in SQL.
- Familiarity with our internal service.

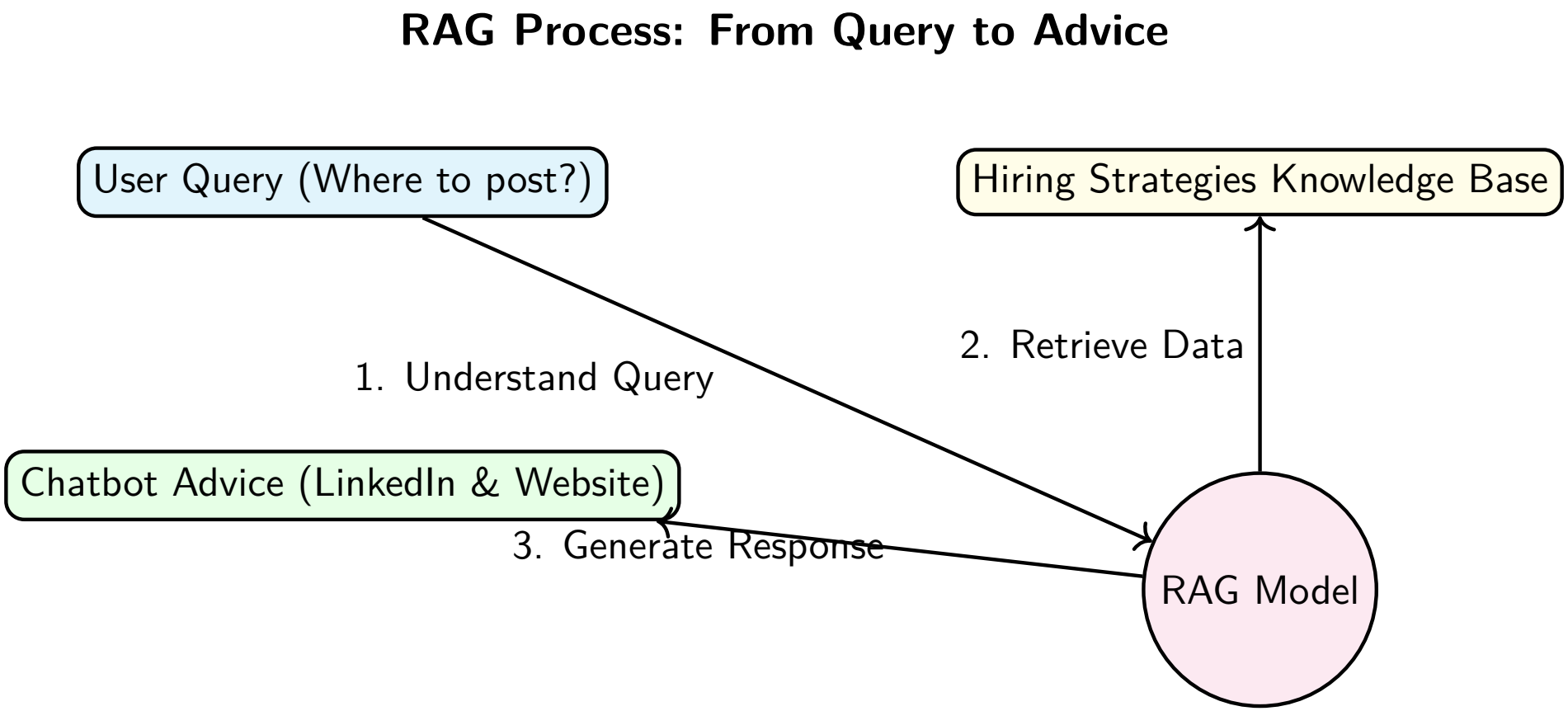
RAG Chatbot V2: Smarter Job Posting with a RAG

User: Where should I post this job description for the best results?

Gemini: Based on our internal hiring strategies, I recommend posting this role on LinkedIn and our company website.

User: That's great advice. Thank you!

RAG Chatbot V2: Smarter Job Posting with a RAG



RAG Chatbot V2: Smarter Shortlisting

User: Okay, I've received a batch of applications. Can you help me shortlist the candidates?

Gemini: Yes. Please share the resumes. I can analyze them against our best practices and required skills to find the most suitable candidates.

User: Great. I've uploaded the applications.

Gemini: Analyzing applications now.

RAG Chatbot V2: Smarter Interview Scheduling

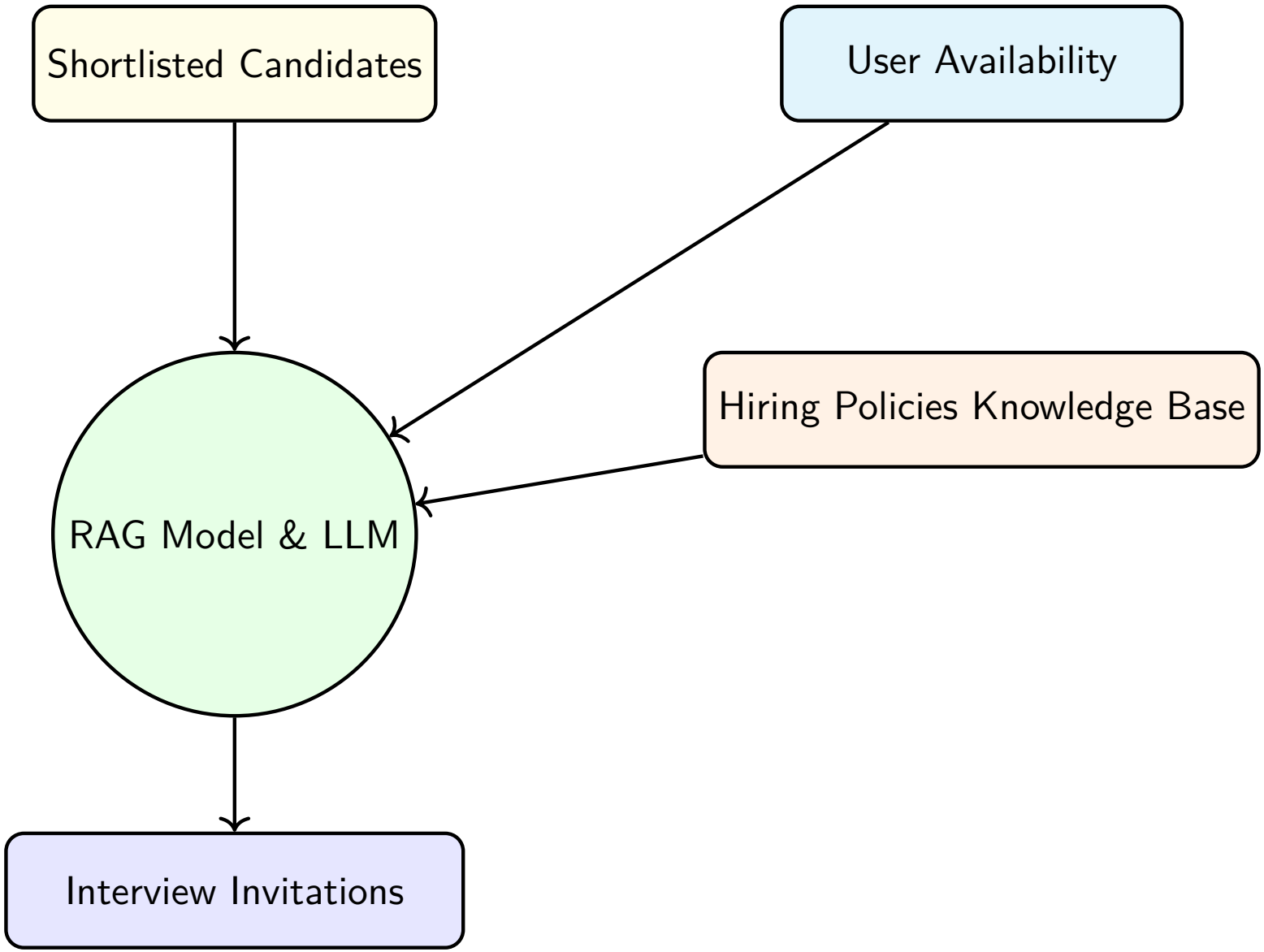
User: Okay, I have a list of shortlisted candidates. Can you help me schedule interviews?

Gemini: Yes. Please share your availability. I will then use our internal policies to draft invitations with suitable time slots.

User: I'm available next Tuesday and Wednesday from 10 am to 4 pm.

Gemini: Understood. I will draft interview invitations for these times.

RAG Chatbot V2:: Smarter Interview Scheduling



RAG Chatbot V2: Generating Interview Questions

User: What kind of questions can I ask for this role?

Gemini: Based on our internal interview question bank, I can generate a list of questions for you. Would you like me to do that?

User: Yes, please generate a list of questions focusing on Python and Django.

RAG Chatbot V2: Generating Interview Questions

From Interview Question Bank for Backend Engineers:

- **Python:** How do you handle asynchronous programming in Python? Provide an example.
- **Django:** Explain the Django MVT (Model-View-Template) architecture. How does it differ from a traditional MVC?
- **Databases:** What is the purpose of migrations in Django?

Problem-Solving:

- Walk me through the process of debugging a Django application that is experiencing slow load times.

RAG Chatbot V2: Drafting the Final Offer Letter

User: We have selected a candidate. Can you please draft the offer letter for Mr. Rahul?

Gemini: Yes. I will use our standard offer letter template to create the document and our internal onboarding checklist to ensure all details are included.

User: The salary is 95,000 Rupees per month with a start date of Oct 1st. Please include those details.

Gemini: Understood. Drafting the offer letter and onboarding checklist now.

RAG Chatbot V2: Automated Onboarding Materials

Gemini: The personalized welcome message and onboarding checklist for Rahul are ready.

User: Perfect. This saves a lot of time. Thank you!

Generated Onboarding Documents

Welcome to the Team, Rahul! Dear Rahul, Welcome to [Company Name]!

We are thrilled to have you join us as a Backend Engineer. We've prepared a checklist to help you get started on your first day, October 1st.

Onboarding Checklist

- Complete HR forms and submit ID
- Review employee handbook
- Get your laptop and workspace set up
- Meet with your team lead, [Manager's Name]

Solved vs. Unsolved Limitations: RAG Chatbot V2

Problem	Status
Reactive, Not Proactive	✗
Lacks Memory	✗
Specific Advice	✓
Can't Take Action	✗

Chatbot V3: Tool-Augmented Chatbot

Chatbot Version 3: Tool-Augmented Chatbot

We've integrated the chatbot with external tools via APIs.

- LinkedIn API
- Mail API
- Calendar API
- HRM Tool
- Resume Parser Tool

Tool-Augmented Chatbot V3: JD Drafting

User: Can you please help me draft a Job Description to hire a Backend Engineer?

Gemini: Yes. I will draft a JD using the company's past templates and high-performing examples to ensure it aligns with our standards.

User: Sounds good. Let's go ahead with that.

- *Note:* No external tools are used in this phase. The chatbot's capability here is based on its internal knowledge (RAG).

Tool-Augmented Chatbot V3: Automated Job Posting

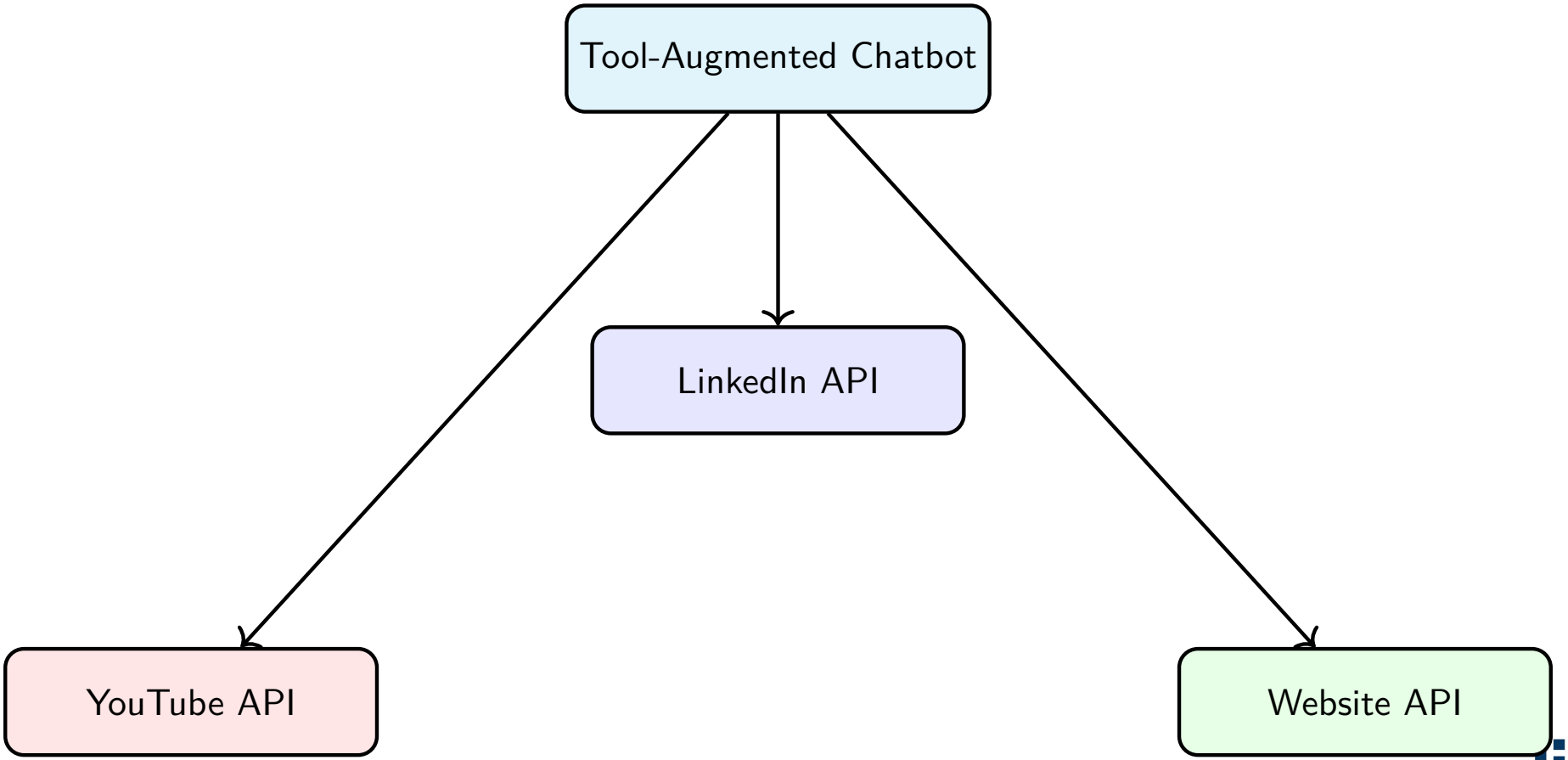
User: The JD is ready. Can you post it on LinkedIn and our company careers page?

Gemini: Understood. I will use the LinkedIn API, YouTube API and our company website API to post the job automatically.

Gemini: Posting JD on LinkedIn, YouTube and our company website.

Workflow: Tool-Augmented Posting

Uses JD, Customizes posts, Automates action



Tool-Augmented Chatbot V3: JD Revision

User: Can you check how many applications we have received?

Gemini: Connecting to LinkedIn, we have received just 2 applications.

User: What can be done?

Gemini: Here is something which has worked in earlier cases: 1. Broaden JD 2. Boost post on LinkedIn.

User: Revise the JD.

Tool-Augmented Chatbot V3: Automated Shortlisting

User: Total 10 applications have been received, can you please help me shortlisting the candidates?

Gemini: Sure, I will use the resume parser tool. I will first download all resumes from LinkedIn.

Gemini: Studying the JD, studying the resumes. 3 candidates are shortlisted.

User: Mail me their profiles.

Gemini: Sending mail.

Tool-Augmented Chatbot V3: Automated Scheduling

User: I have selected 2 candidates. Can you draft a mail and schedule interviews?

Gemini: Sure, let me check for your availability this week. You are free this Friday. Should I schedule the interview?

User: Yes, that works.

Gemini: Drafting a mail. Mail is ready. Sending.

Tool-Augmented Chatbot V3: Generating Interview Questions

User: What kind of questions can I ask in the interview?

Gemini: Here are common questions asked in past interviews: 1. Describe a backend system you designed 2. How to design a rate limiter 3. How to scale a system for high traffic.

User: Can you generate a list of questions?

Generated Questions:

1. Describe a backend system you designed
2. How to design a rate limiter
3. How to scale a system for high traffic

Tool-Augmented Chatbot V3: Draft Offer Letter

User: I have selected 1 candidate. Can you draft and send the offer letter?

Gemini: Sure, here is a sample offer letter. Please review it.

User: Yes, this works.

Gemini: Sending the mail.

Tool-Augmented Chatbot V3: Automated Onboarding

User: The candidate has accepted the offer. Can you send the welcome email?

Gemini: Sure, I have drafted the welcome email. Sending.

User: Can you also trigger the onboarding process?

Gemini: Sure, I will execute the following tasks in the HRMS:

1. Share employment contract
2. Create official email ID
3. Provision hardware
4. Schedule KT session

Solved vs. Unsolved Status after Tool-Augmented V3 Chatbot

Problem	Status
Reactive, Not Proactive	×
Lacks Memory	×
Specific Advice	✓
Can take Action	✓
Can't adapt	×

Chatbot V4: Agentic Chatbot

Chatbot Version 4: The Agentic Chatbot

Problem: The previous chatbots were reactive and lacked a holistic view of the hiring process. They required step-by-step instructions from the user.
Solution: By integrating planning capabilities and persistent memory, this version can act as a true agent. It takes a high-level goal and generates a complete, autonomous plan to achieve it.

User: I want to hire a backend engineer.

Gemini: Goal: Hire a remote backend engineer (2-4 years experience). Proposed plan:

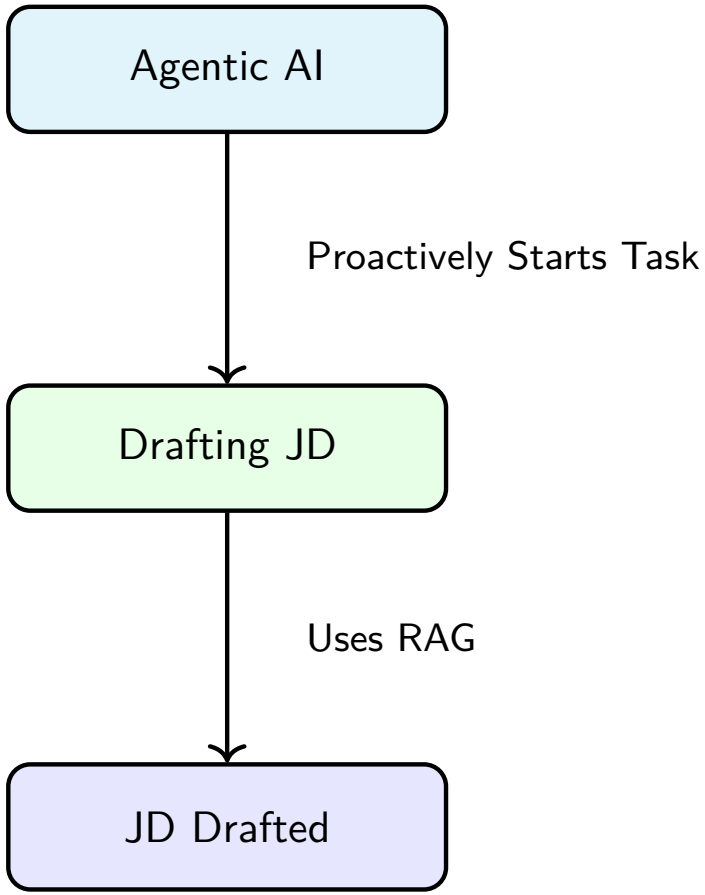
1. Draft JD → post on best platforms
2. Monitor pipeline → adjust strategy if needed
3. Screen candidates → schedule interviews
4. Draft + send offer → start onboarding

Agentic Chatbot V4: JD Drafting (The First Step)

User: I want to hire a backend engineer.

Gemini: Understood. I am starting the first task in our plan: drafting the JD using our internal templates. Do you want any Change?

User: No. This is fine.

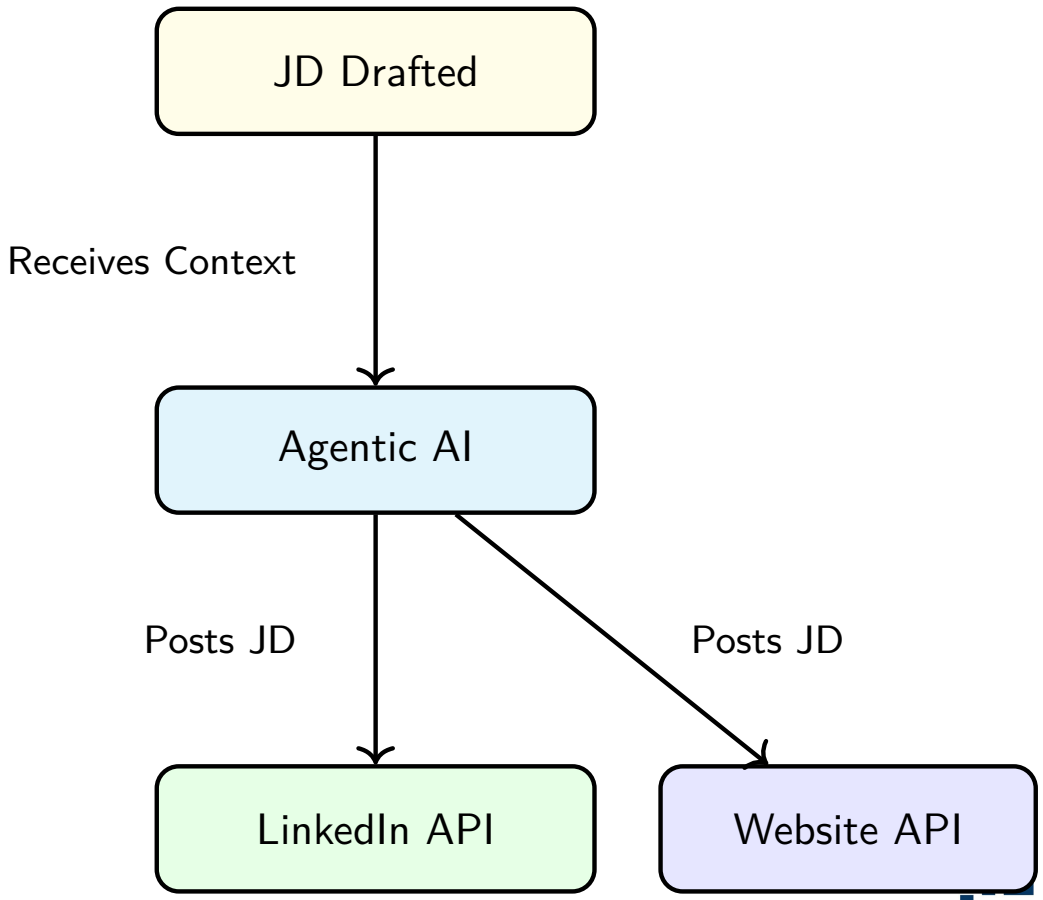


Agentic Chatbot V4: Automated JD Posting (Second Step)

Gemini: I am now posting it on LinkedIn and our company careers page as per our plan. Shall I go ahead?

User: Yes, Go ahead.

Gemini: Job Posted. I will monitor the application and keep you posted.



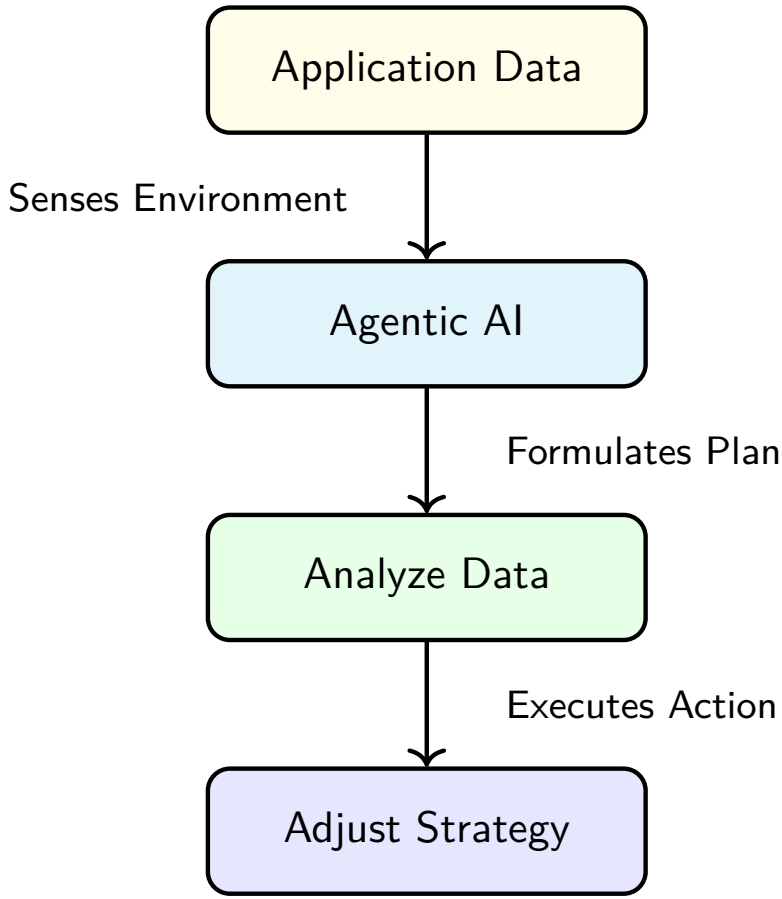
Agentic Chatbot V4: Proactively Adjusting Strategy

Gemini: The JD has been live for 48 hours. I'm monitoring the pipeline and notice we've only received 3 applications.

Gemini: I'm going to revise the JD to broaden the requirements and then boost the post on LinkedIn to attract more candidates. Shall I go ahead?

User: Yes, Please.

Gemini: Revised JD Posted. I will continue monitoring.

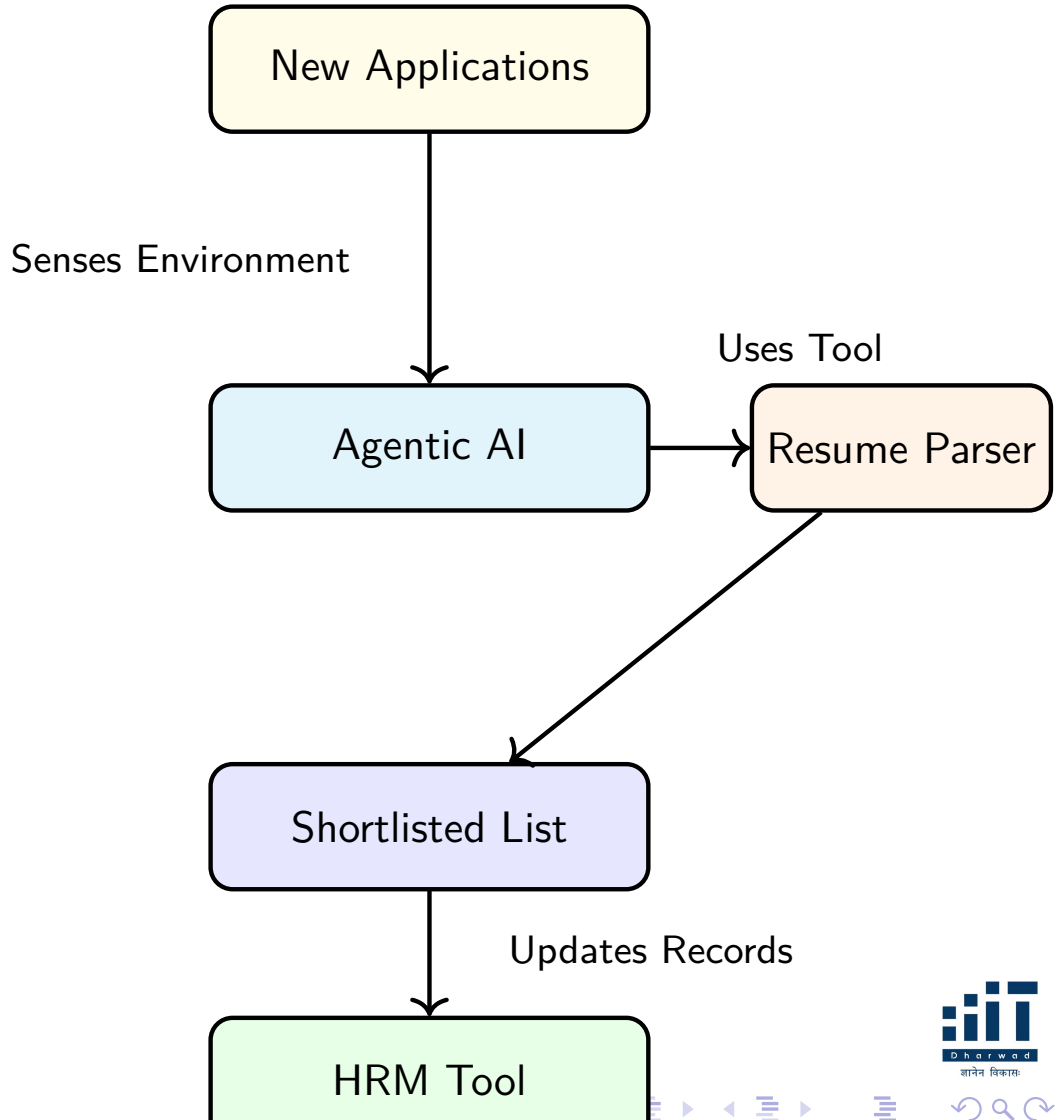


Agentic Chatbot V4: Automated Shortlisting and Screening

Gemini: 7 applications received. I am now automatically screening all new applications against our JD.

Gemini: I have finished screening. The profiles of the 5 most qualified candidates have been uploaded to the HRM tool. Shall I schedule the interview?

User: Go ahead.

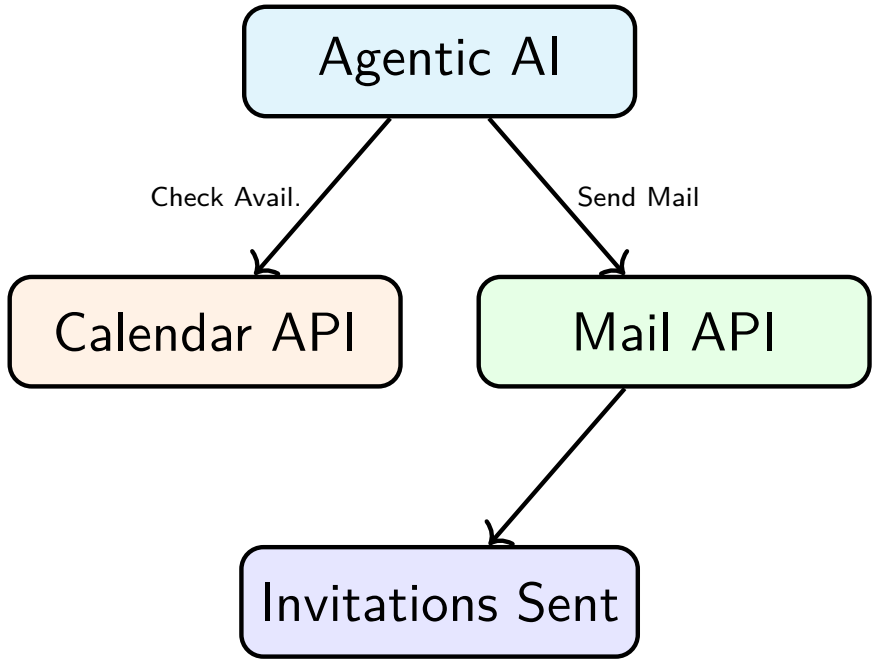


Agentic Chatbot V4: Automated Scheduling

Gemini: I have checked your calendar. It says you are free on Friday. Shall I schedule it on Friday?

User: Yes.

Gemini: Scheduling. Drafting email. Email sent.



Agentic Chatbot V4: Proactive Interview Assistance

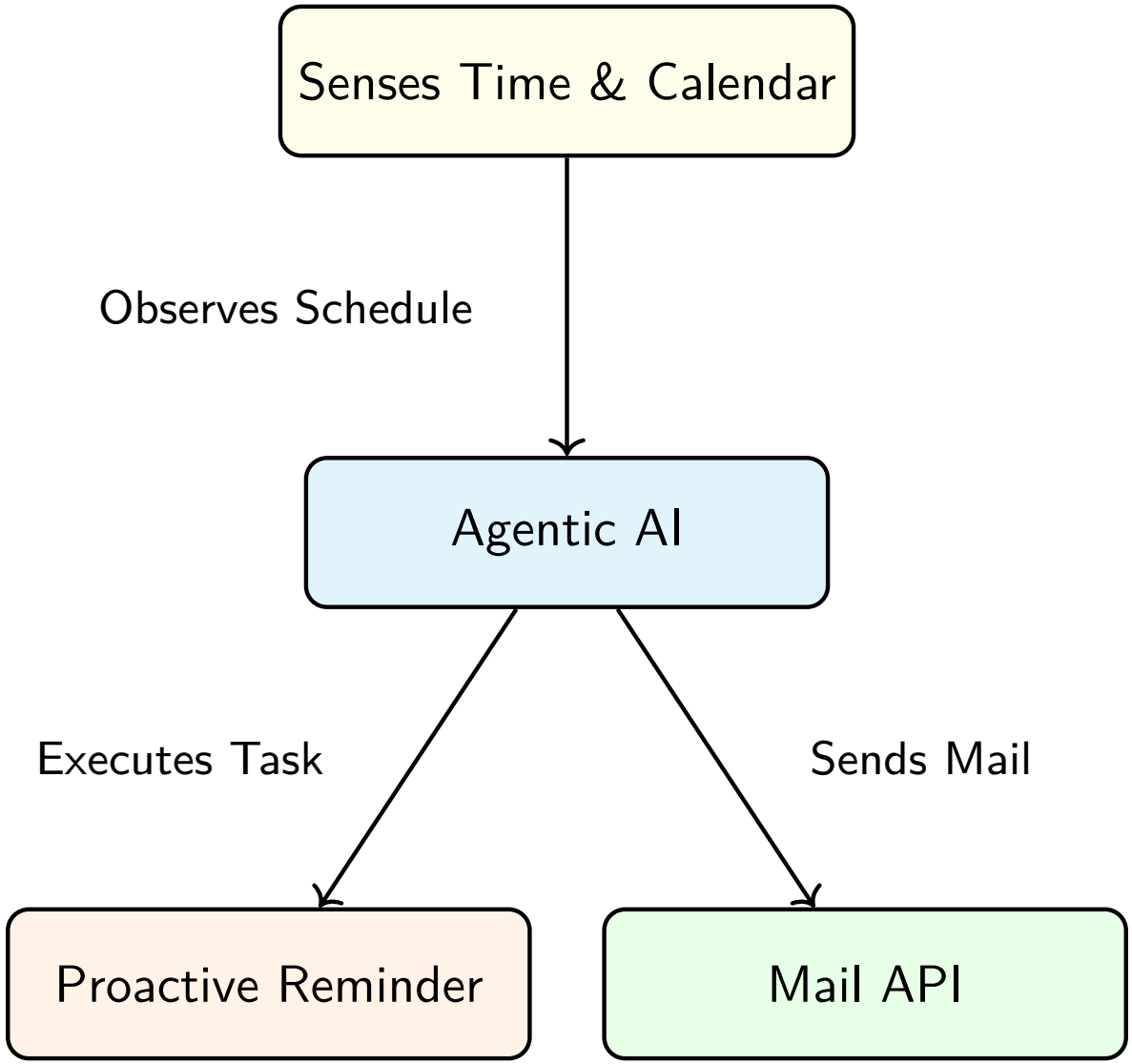
Gemini: Quick reminder! You have 2 interviews lined up for today.

User: Thanks for reminding.

Gemini: I have mailed you a document containing a list of interview questions asked in previous interviews for the same role.

User: Great, I will check it out.

Workflow: Proactive Interview Assistance



Agentic Chatbot V4: Drafting and Sending Offer

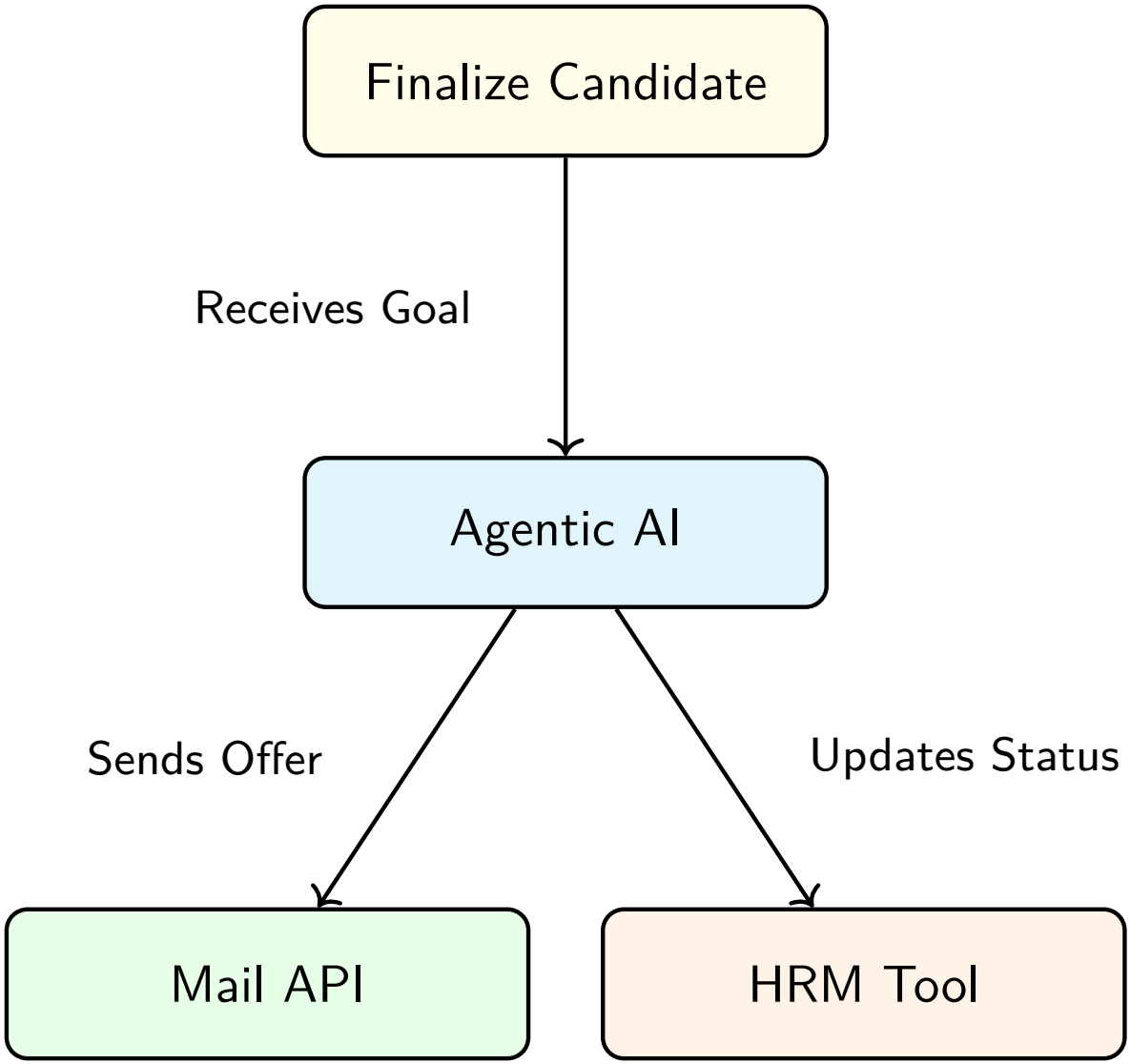
User: I have finalized 1 candidate. Can you draft an offer letter?

Gemini: Sure, here is an offer letter. Please review.

User: Yes this works.

Gemini: Offer letter sent. Tracking acceptance.

Workflow: Proactive Offer Management



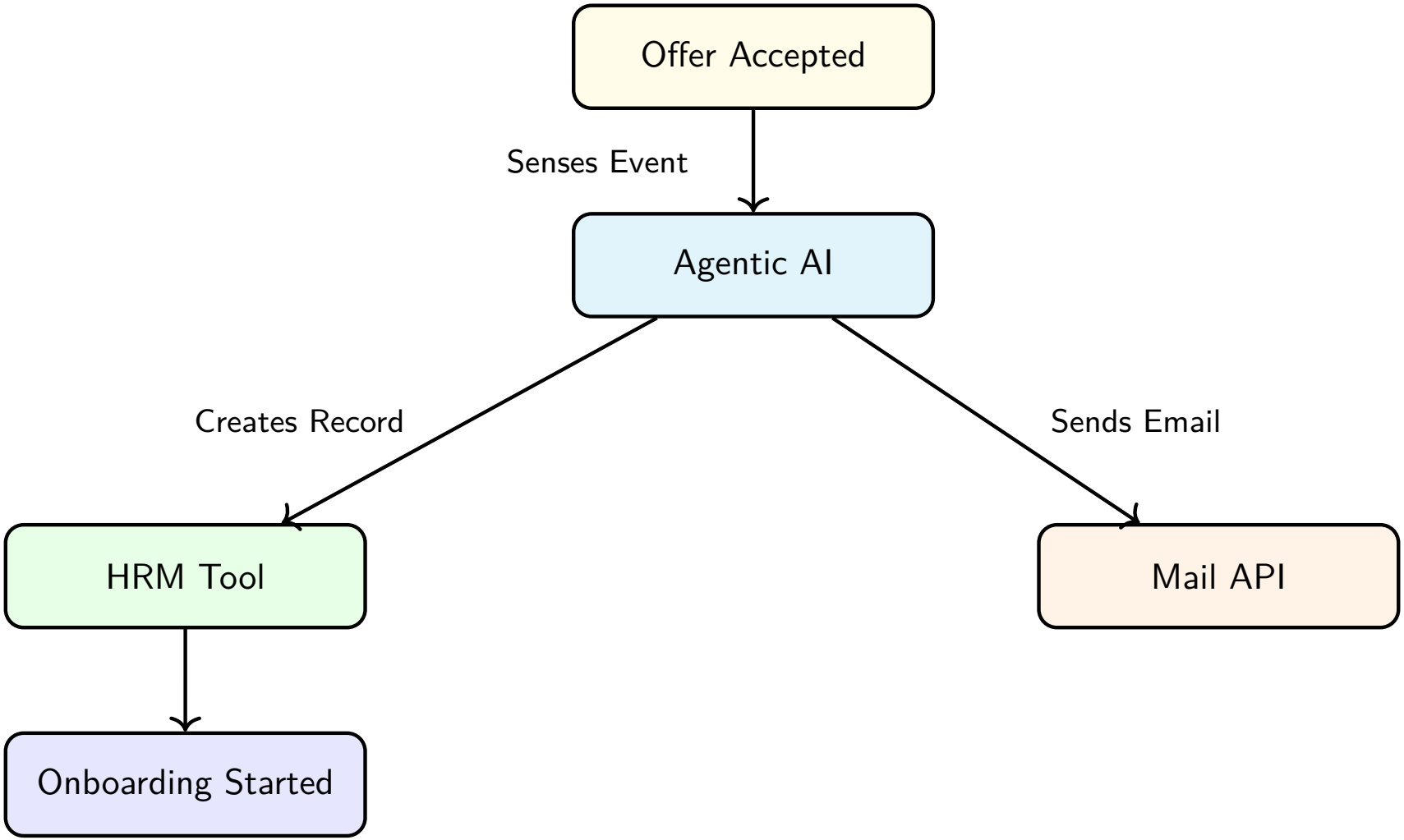
Agentic Chatbot V4: Automated Onboarding

Gemini: The candidate has accepted the offer. I am now automatically triggering the onboarding process.

Gemini: I will send the welcome email and create the employee record in our HRM tool. Shall I schedule intro meeting?

User: Yes, Please.

Workflow: Proactive Onboarding



Solved vs. Unsolved Status after V4 Agentic Chatbot

Problem	Status
Reactive, Not Proactive	✓
Lacks Memory	✓
Specific Advice	✓
Can take Action	✓
Can't adapt	✓

Chatbot Comparison: From Reactive to Agentic

Chatbot Version	Proactive	Memory	Grounded Advice	Takes Action	Adaptability
V1: Generic	✗	✗	✗	✗	✗
V2: RAG-Based	✗	✗	✓	✗	✗
V3: Tool-Augmented	✗	✗	✓	✓	✗
V4: Agentic AI	✓	✓	✓	✓	✓

Generative AI vs Agentic AI

- **Generative AI** is about **creating content**, while **Agentic AI** is about **solving a goal**.
- **Generative AI** is typically **reactive**, responding to prompts. **Agentic AI** is **proactive**, autonomously taking action to achieve a goal.
- **Generative AI** serves as a foundational **building block** for an Agentic AI system, providing the core capability to generate new text or code.

Generative AI is a capability; Agentic AI is a behavior.

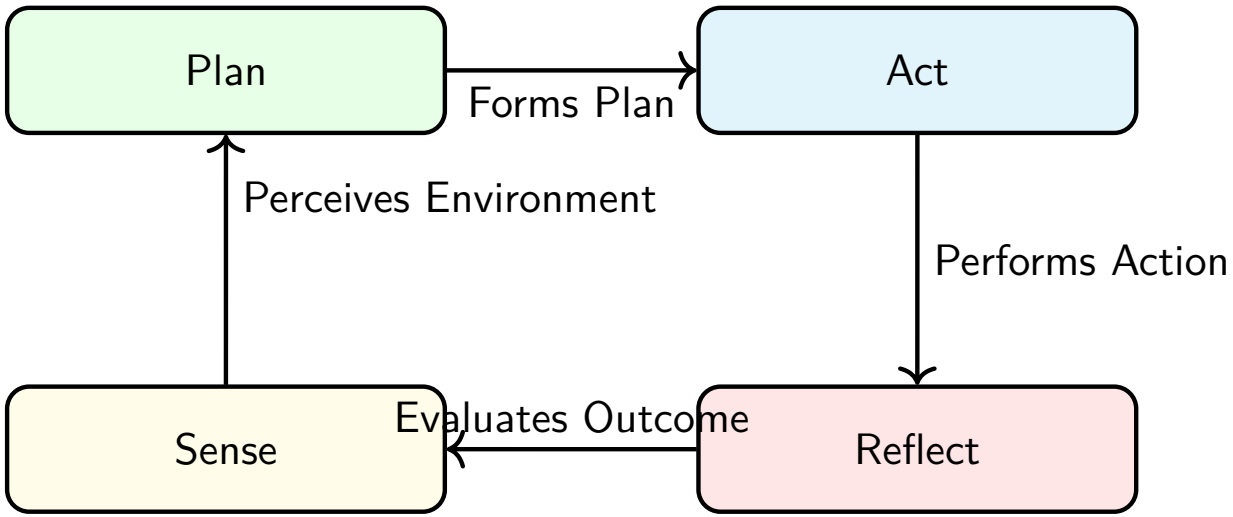
Defining Agentic AI

What is Agentic AI?

Agentic AI is an autonomous system designed to achieve a specific, high-level goal. Unlike a reactive chatbot, it is a proactive entity that can operate independently with minimal human inetervention.

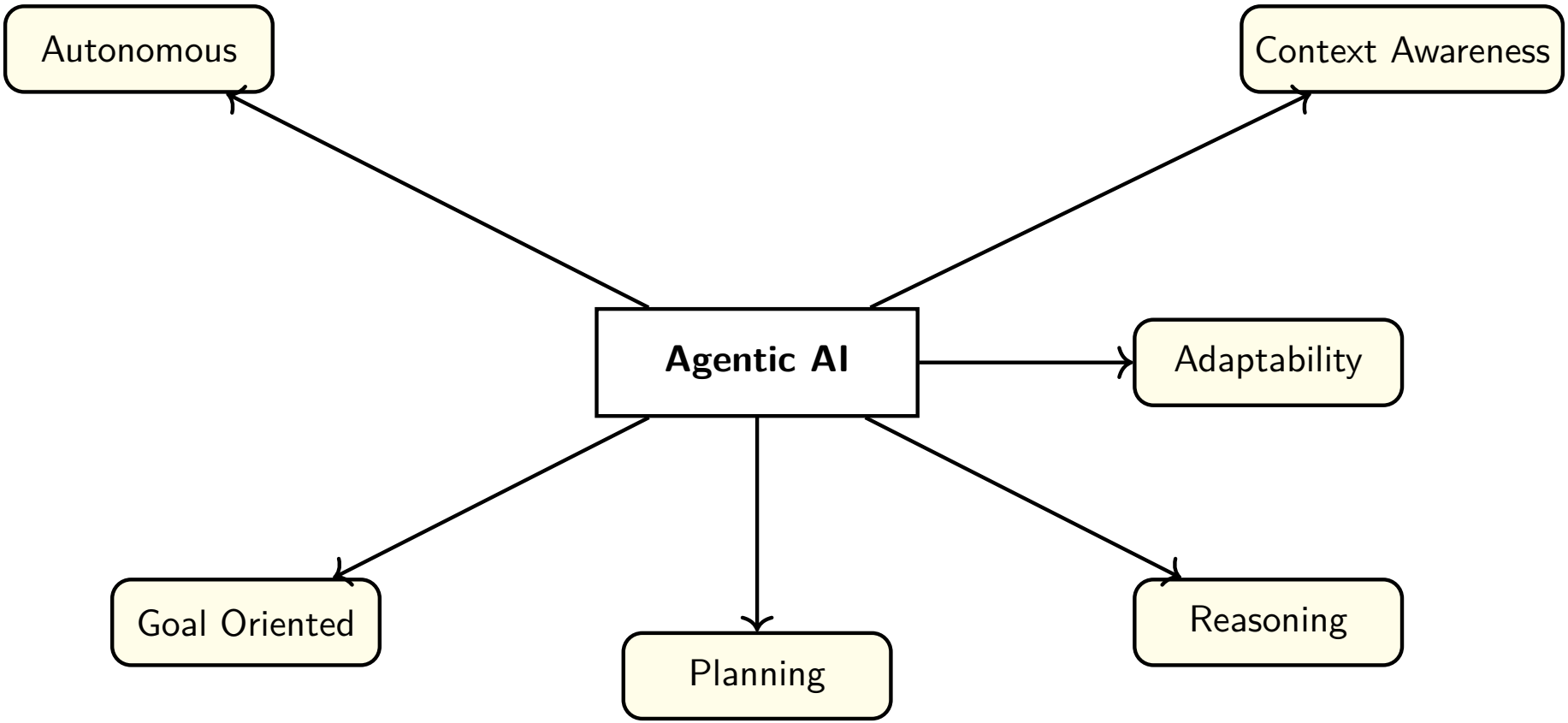
Its core capabilities include:

- **Planning:** Breaks down a complex goal into a series of smaller steps.
- **Reasoning:** Determines the best course of action and the tools needed.
- **Memory:** Retains context from past actions and conversations.
- **Tool Use:** Interacts with external systems (APIs, databases, etc.).



Agentic AI Characteristics

Key Characteristics of Agentic AI



Key Characteristics of Agentic AI: Explained

- **Autonomous:** *Acts independently* to pursue its goals without requiring constant human intervention. It can initiate actions on its own.
- **Goal-Oriented:** Designed to *achieve specific, predefined objectives*. Its entire operation revolves around progressing towards these goals.
- **Planning:** Capable of *breaking down complex goals into a sequence of actionable steps*. This includes anticipating outcomes and strategizing.
- **Reasoning:** Possesses the ability to *make logical inferences and decisions* based on available information, past experiences, and its current context.
- **Context Awareness:** Maintains an understanding of its *environment, past interactions, and current state*. This enables more relevant and informed actions.
- **Adaptability:** Can *adjust its plans and behaviors* in response to new information, unexpected changes in the environment, or failures in execution.

Autonomy

- **Definition:** Autonomy is the AI's ability to make decisions and take actions on its own to achieve a goal, without needing step-by-step human instructions. It is proactive, not just reactive.
- **Facets of Autonomy:**
  - **Execution:** The AI performs a task, such as posting a job or sending an email.
  - **Decision Making:** The AI chooses which action to take based on its goal and current state.
  - **Tool Usage:** The AI selects and uses the appropriate tools (APIs) to complete a task.
- **How Autonomy is Controlled:**
  - **Human-in-the-Loop (HITL):** Checkpoints where human approval is required before the AI proceeds.
  - **Permission Scope:** Limiting which tools or actions the AI is allowed to perform independently.
  - **Guardrails:** Defining hard rules or ethical boundaries that the AI must follow.
- **Potential Dangers:**
  - Without proper controls, autonomous agents can send incorrect offers, shortlist candidates with bias, or misuse company resources.

Goal-Oriented

Being goal-oriented means that the AI system operates with a persistent objective in mind and continuously directs its actions to achieve that objective, rather than just responding to isolated prompts.

Key Characteristics:

- Goals serve as a *compass* for autonomy
- Goals can be bound by explicit *constraints*
- Goals are encoded in the system's *core memory*
- Goals are *dynamic* and can be revised

Example: Core Memory Representation

```
{   "main_goal": "Hire a backend
engineer",
  "constraints": {
    "experience": "2-4 years",
    "remote": true,
    "stack": ["Python", "Django"]
},
  "status": "active",
  "created_at": "2025-09-04",
  "progress": {
    "JD_created": true,
    "posted_on": "LinkedIn",
    "applications_received": 8,
    "interviews_scheduled": 2   } }
```

Planning

Agent's ability to break down a high-level goal into a structured sequence of actions or subgoals and decide the best path to achieve the desired outcome.

Step 1: Generating multiple candidate plans

- Plan A: Post JD on LinkedIn, GitHub Jobs, AngelList
- Plan B: Use internal referrals and hiring agencies

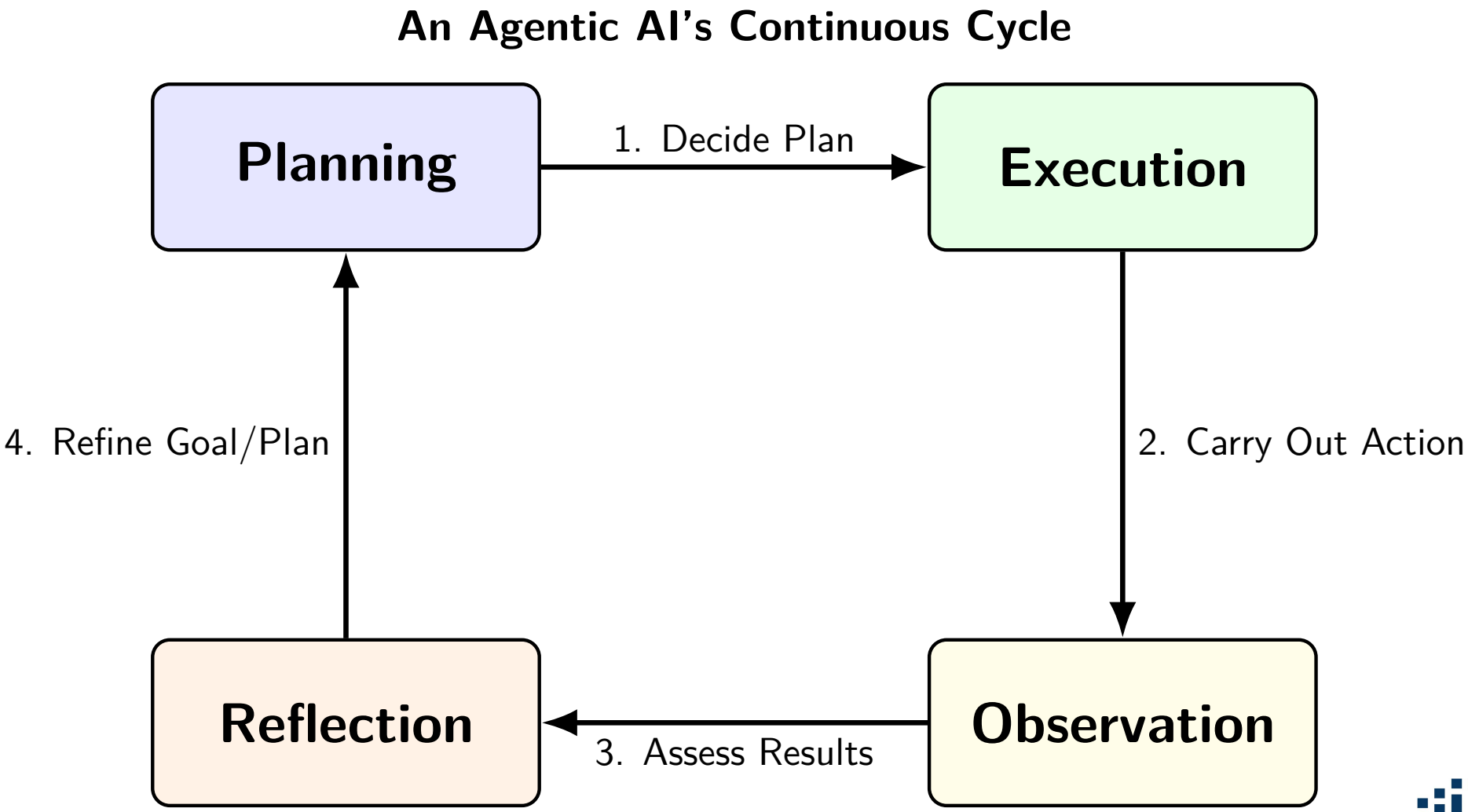
Step 2: Evaluate each plan

- **Efficiency** (Which is faster?)
- **Tool Availability** (Which tools are available)
- **Cost** (Does it require premium tools?)
- **Risk** (Will it fail if we get no applicants?)
- **Alignment with constraints** (remote-only? budget?)

Step 3: Select the best plan with the help of:

- Human-in-the-loop input (e.g., "Which of these options do you prefer?")
- A pre-programmed policy (e.g., "Favor low-cost channels first")

The Planning and Execution Loop



Reasoning

Reasoning is the cognitive process through which an agentic AI system interprets information, draws conclusions, and makes decisions - both while planning ahead and while executing actions in real time.

Reasoning During Planning:

1. **Goal decomposition** - Break down abstract goals into concrete steps
2. **Tool selection** - Decide which tools will be needed for which steps
3. **Resource estimation** - Estimate time, dependencies, risks

Reasoning During Execution:

1. **Decision-making** - Choosing between options (3 candidates match $\rightarrow$ schedule 2 best, reject 1)
2. **HITL handling** - Knowing when to pause and ask for help (Unsure about salary range)
3. **Error handling** - Interpreting tool/API failures and recovering

Adaptability

Adaptability is the agent's ability to modify its plans, strategies, or actions in response to unexpected conditions — all while staying aligned with the goal.

1. **Failures** (Calendar API)
2. **External feedback** (Less no of applications)
3. **Changing goals** (Hiring a freelancer)

Context Awareness

Context awareness is the agent's ability to understand, retain, and utilize relevant information from the ongoing task, past interactions, user preferences, and environmental cues to make better decisions throughout a multi-step process.

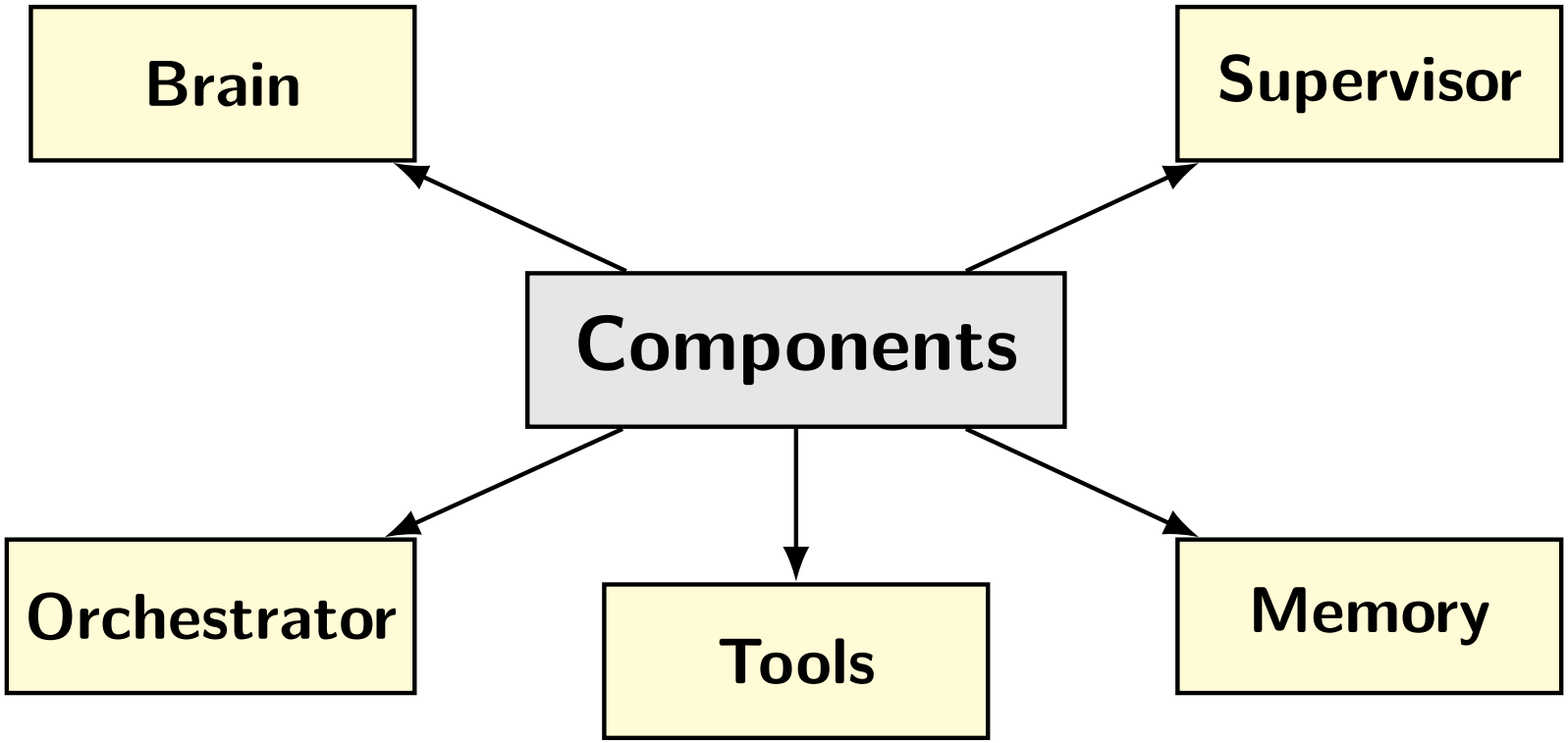
1. Types of context
   a. The original goal
   b. Progress till now + Interaction history (**Job description was finalized and posted to LinkedIn & GitHub Jobs**)
   c. Environment state (Number of applicants so far = 8 or LinkedIn promotion ends in 2 days)
   d. Tool responses (Resume parser → "Candidate B has 3 years Django + AWS experience or Calendar API → "No conflicts at 2 PM Wednesday)
   e. User specific preferences (Prefers remote-first candidates or Likes receiving interview questions in a Google Doc)
   f. Policy or Guardrails (Do not send offer without explicit user approval or Never use platforms that require paid ads unless approved)
2. Context awareness is implemented through memory
3. Short term memory: Current Session
4. Long term memory: since inception

Key Characteristics of Agentic AI- Summarized

- **Goal Oriented**
  The agent operates with a persistent objective in mind, directing all actions toward achieving that goal.
- **Planning**
  The ability to break down a high-level goal into a structured sequence of actions and subgoals.
- **Reasoning**
  The cognitive process of interpreting information, making decisions, and drawing conclusions during both planning and execution.
- **Adaptability**
  The capacity to modify plans and actions in real time in response to unexpected conditions or failures.
- **Context Awareness**
  The ability to understand and utilize relevant information from past interactions, user preferences, and the environment to make informed decisions.

Agentic AI Components

Components of an Agentic AI System



- **Brain:** Large Language Model (LLM) that acts as the reasoning engine.
- **Orchestrator:** Breaks down high-level goals into executable sub-tasks.
- **Tools:** External functions/APIs the agent can use to interact with the world.
- **Memory:** Contextual information for long-term and short-term use.
- **Supervisor:** Monitors the agent's actions to ensure it stays on track and adheres to safety policies.

The Brain: The Core of the Agent

Goal Interpretation	Understands user instructions and translates them into objectives.
Planning	Breaks down high-level goals into subgoals and ordered steps.
Reasoning	Makes decisions, resolves ambiguity, and evaluates trade-offs.
Tool Selection	Chooses which tool(s) to use at a given step.
Communication	Generates natural language outputs for humans or other agents.

The Orchestrator: Nervous System or Manager

Task Sequencing	Determines the order of actions (step 1 → step 2 → ...).
Conditional Routing	Directs flow based on context (e.g., if failure, retry or escalate).
Retry Logic	Handles failed tool calls or reasoning attempts with backoff.
Looping & Iteration	Repeats steps (e.g., keep checking job apps until 10 are received).
Delegation	Decides whether to hand off work to tools, LLM, or human.

Tools: The Agent's Hands and Eyes

External Actions	Perform API calls (e.g., post a job, send an email, trigger onboarding).
Knowledge Base Access	Retrieve factual or domain-specific information using RAG or search tools to ground responses.

Memory: An Agent's Awareness

Short-Term Memory	Maintains the active session's context — recent user messages, tool calls, and immediate decisions.
Long-Term Memory	Persists high-level goals, past interactions, user preferences, and decisions across sessions.
State Tracking	Monitors progress: what's completed, what's pending (e.g., "JD posted", "Offer sent").

The Supervisor: Ensuring Safety and Control

Approval Requests (HITL)	Agent checks with human before high-risk actions (e.g., sending offers).
Guardrails Enforcement	Blocks unsafe or non-compliant behavior.
Edge Case Escalation	Alerts humans when uncertainty/conflict arises.

Thank You!